

A FORMALLY VERIFIED SPARSE FOCK CONSTRUCTION FOR ERDŐS PROBLEM 906

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ABSTRACT. I study the explicit entire function

$$F(z) = \sum_{j \geq 1} \frac{z^{\lfloor j^{3/2} \rfloor}}{\sqrt{\lfloor j^{3/2} \rfloor!}}.$$

Every nonempty open subset of the plane contains a zero of $F^{(n)}$ for every sufficiently large n . On each fixed annulus the zeros admit a quantitative description: they lie in pairwise disjoint disks around explicit two-term model points, each such disk contains exactly one simple zero, and the zero set has covering radius $O(n^{-1/6})$. The construction is deterministic. Every theorem, proposition, and corollary stated in this note has a corresponding declaration in the accompanying Lean 4/Mathlib development. No claim of being the first solution of Erdős Problem 906 is made.

1. STATEMENT AND CONTEXT

For a nonzero entire function g , write $\mathcal{Z}(g)$ for its zero set. The cofinite zero-hitting property is

$$\forall U \subset \mathbb{C} \text{ nonempty open, } \exists N(U) \quad \forall n \geq N(U) : \quad \mathcal{Z}(g^{(n)}) \cap U \neq \emptyset. \quad (1.1)$$

For a transcendental entire function this is equivalent to the formulation of Erdős Problem 906 using every strictly increasing sequence of derivative orders. Indeed, if (1.1) fails for an open set U , the infinitely many exceptional orders contain a strictly increasing sequence whose derivative-zero union misses U . Conversely, a cofinite set of orders meets every infinite increasing sequence eventually.

Erdős recorded the problem in [2, p. 72, IV.1(i)]. Barth and Schneider [1, Theorem 1] solved a different interpolation problem: for prescribed discrete sets S_k one may choose derivative orders n_k so that $f^{(n_k)}$ vanishes on S_k . That result does not contain the cofinite quantifier in (1.1). Gethner’s final-set framework likewise concerns points met by zeros of infinitely many derivatives, a weaker quantifier than (1.1) [3]. Hou recently gave a probabilistic bounded-coefficient Fock-series construction satisfying (1.1), together with a Lean formalization of his main theorem and growth bound [4]. The point here is different: the function below is explicit and sparse, and the derivative zeros can be localized quantitatively. I make no historical-priority claim beyond these stated features.

Put

$$\nu_j = \lfloor j^{3/2} \rfloor, \quad q_j = \nu_{j+1} - \nu_j, \quad F(z) = \sum_{j=1}^{\infty} \frac{z^{\nu_j}}{\sqrt{\nu_j!}}. \quad (1.2)$$

For $0 < a \leq b$, set

$$A_{a,b} = \{z \in \mathbb{C} : a \leq |z| \leq b\}.$$

For an active index j with $\nu_j \geq n$, define

$$m_j = \nu_j - n, \quad A_j = \frac{\sqrt{\nu_j!}}{m_j!}, \quad T_j(z) = A_j z^{m_j}, \quad \rho_j = \left(\frac{A_j}{A_{j+1}} \right)^{1/q_j}. \quad (1.3)$$

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The radius ρ_j is where $|T_j| = |T_{j+1}|$. The corresponding nonzero zeros of the two-term model $T_j + T_{j+1}$ are

$$z_{j,\ell} = \rho_j \exp\left(\frac{(2\ell+1)\pi i}{q_j}\right), \quad 0 \leq \ell < q_j. \quad (1.4)$$

For $c > 0$ write

$$\mathcal{D}_{j,\ell}(c) = \overline{D}\left(z_{j,\ell}, \frac{c\rho_j}{q_j}\right). \quad (1.5)$$

Theorem 1.1 (Cofinite derivative zeros). *For every nonempty open set $U \subset \mathbb{C}$ there is N such that, for every $n \geq N$, the derivative $F^{(n)}$ has a zero in U .*

Theorem 1.2 (Quantitative annular covering). *Let $0 < a \leq b$. There are constants $C > 0$ and N such that for every $n \geq N$ and every $w \in A_{a,b}$ there is a zero z of $F^{(n)}$ satisfying*

$$|z - w| \leq Cn^{-1/6}. \quad (1.6)$$

Proposition 1.3 (One simple zero in each model disk). *Let $0 < a \leq b$. There exist $c \in (0, 1]$ and N such that for every $n \geq N$, every active j with $a \leq \rho_j \leq b$, and every $0 \leq \ell < q_j$, the disk $\mathcal{D}_{j,\ell}(c)$ contains exactly one zero of $F^{(n)}$. That zero is simple.*

Proposition 1.4 (Exhaustion of annular zeros). *Let $0 < a \leq b$ and $c > 0$. For all sufficiently large n , every zero z of $F^{(n)}$ with $a \leq |z| \leq b$ belongs to a disk $\mathcal{D}_{j,\ell}(c)$ with*

$$\frac{a}{2} \leq \rho_j \leq 2b, \quad 0 \leq \ell < q_j.$$

Corollary 1.5 (Eventual annular simplicity). *Let $0 < a \leq b$. For all sufficiently large n , every zero of $F^{(n)}$ in $A_{a,b}$ is simple.*

Proposition 1.6 (Pairwise disjoint model disks). *Let $0 < a \leq b$. If $0 < c \leq 1$, then for all sufficiently large n the admissible disks $\mathcal{D}_{j,\ell}(c)$ with $a \leq \rho_j \leq b$ are pairwise disjoint.*

Proposition 1.7 (Finite model-disk zero count). *Let $0 < a \leq b$. There exist $c \in (0, 1]$ and N such that, for every $n \geq N$ and every finite set I of admissible pairs (j, ℓ) with $a \leq \rho_j \leq b$, the union of the disks indexed by I contains exactly $|I|$ distinct zeros of $F^{(n)}$, all simple.*

The remainder of the paper proves these statements in a form that mirrors the structure of the formal development. The exact order and type, sectorial zero asymptotics, limiting zero measures, other values of the support exponent, and the critical endpoint $p = 4/3$ are deliberately not part of this submission.

2. SPARSE SUPPORT AND DERIVATIVE WEIGHTS

The two relevant scales are immediate from the support. The mean-value theorem and the floor error give

$$\frac{3}{2}\sqrt{j} - 1 \leq q_j \leq \frac{3}{2}(\sqrt{j} + 1) + 1 \quad (j \geq 1). \quad (2.1)$$

Thus, when $\nu_j = n + O(\sqrt{n})$,

$$j \asymp n^{2/3}, \quad q_j \asymp n^{1/3}. \quad (2.2)$$

In particular the first active exponent m_{j_0} , where $j_0 = \min\{j : \nu_j \geq n\}$, is $O(n^{1/3})$.

The series in (1.2) is entire. A convenient uniform majorant is

$$|F(z)| \leq \sqrt{1 + |z|^2} \exp\left(\frac{1 + |z|^2}{2}\right). \quad (2.3)$$

Indeed, the sparse series is termwise dominated by $\sum_{k \geq 0} |z|^k / \sqrt{k!}$, and a Cauchy–Schwarz estimate gives (2.3). Termwise differentiation is therefore legitimate, and

$$F^{(n)}(z) = \sum_{\nu_j \geq n} A_j z^{m_j}. \quad (2.4)$$

The support is infinite, hence F is not a polynomial. In the Lean development transcendence is certified directly by showing that sufficiently high derivatives of a polynomial vanish while the corresponding derivatives of F do not.

For integers $k \geq n$ and $r > 0$, set

$$\Phi_{n,r}(k) = \frac{1}{2} \log(k!) - \log((k-n)!) + (k-n) \log r. \quad (2.5)$$

Its first difference is

$$d_{n,r}(k) = \Phi_{n,r}(k+1) - \Phi_{n,r}(k) = \log r + \frac{1}{2} \log(k+1) - \log(k+1-n). \quad (2.6)$$

Writing $m = k - n$,

$$d_{n,r}(k) - d_{n,r}(k+1) = \log\left(1 + \frac{1}{m+1}\right) - \frac{1}{2} \log\left(1 + \frac{1}{n+m+1}\right) > 0. \quad (2.7)$$

Hence $\Phi_{n,r}$ is strictly discretely concave. Uniformly for $m = O(\sqrt{n})$, the drop in (2.7) is comparable to $n^{-1/2}$.

The crossing radius has the exact representation

$$\log \rho_j = \frac{1}{q_j} \sum_{s=1}^{q_j} \left\{ \log(m_j + s) - \frac{1}{2} \log(n + m_j + s) \right\}. \quad (2.8)$$

The summand is increasing in s , so the ρ_j are strictly increasing. In the saddle range $m_j = O(\sqrt{n})$, (2.8) and (2.2) give

$$\rho_j = \frac{m_j}{\sqrt{n}} + O(n^{-1/6}), \quad (2.9)$$

and on every fixed compact annulus

$$c_{a,b} n^{-1/6} \leq \rho_{j+1} - \rho_j \leq C_{a,b} n^{-1/6} \quad (2.10)$$

whenever both radii lie in a fixed enlargement of the annulus. These estimates explain the exponent in Theorem 1.2: the radial mesh is $n^{-1/6}$, while the angular mesh on a crossing circle is $q_j^{-1} = O(n^{-1/3})$.

3. TWO-TERM LOCALIZATION

Fix a compact saddle range $m_j \leq B\sqrt{n}$. At $r = \rho_j$ the q_j consecutive differences

$$d_{n,r}(\nu_j), \dots, d_{n,r}(\nu_j + q_j - 1)$$

have sum zero. By (2.7), their endpoint values have opposite signs and magnitude $\gg q_j/\sqrt{n}$. If

$$|q_j \log(|z|/\rho_j)| \leq L,$$

the same signs persist for large n . Since every further sparse step has length $\gg n^{1/3}$, one gains a logarithmic loss

$$E_n \asymp \frac{q_j^2}{\sqrt{n}} \asymp n^{1/6} \quad (3.1)$$

on each step away from the pair $j, j+1$. Summing the resulting geometric tails yields

$$\frac{\sum_{h \neq j, j+1} |T_h(z)|}{|T_j(z)|} \leq (1 + e^L) \frac{e^{-E_n}}{1 - e^{-E_n}}. \quad (3.2)$$

This is the quantitative reduction to two adjacent terms.

Let $z_0 = z_{j,\ell}$ and $q = q_j$. In the logarithmic coordinate $z = z_0 e^w$, division by the nonvanishing principal monomial puts the derivative in the form

$$\Psi(w) = 1 - e^{qw} + E(w). \quad (3.3)$$

Fix a small $c_0 > 0$ and put $\zeta = c_0/q$. The tail estimate (3.2), applied on $|w| \leq \zeta$, gives $|E(w)| \leq K_n$ with $K_n = O(e^{-E_n})$. At the centre $|\Psi(0)| \leq K_n$. On $|w| = \zeta$, the elementary estimate for the exponential remainder gives

$$|1 - e^{qw}| \geq c_0/2$$

when $c_0 \leq 1/2$. Hence $|\Psi| \geq c_0/2 - K_n > K_n$ on the boundary for large n . If Ψ had no zero in $|w| \leq \zeta$, the maximum-modulus principle applied to $1/\Psi$ would force the modulus at the centre to be at least the boundary minimum, a contradiction. Thus a true zero lies within $O(\rho_j/q_j)$ of z_0 .

It remains to prove uniqueness and simplicity without a zero-counting black box. Write $t = z/z_0$ and normalize once more:

$$G(t) = 1 - t^q + R(t). \quad (3.4)$$

For a fixed sufficiently small $s \asymp q^{-1}$, the same tail estimate holds on $|t - 1| \leq 2s$ with $|R(t)| \leq K_n$. Cauchy's estimate therefore gives

$$|R'(t)| \leq K_n/s \quad (|t - 1| \leq s). \quad (3.5)$$

Choose s so that $qs \leq 1/8$. Then t^{q-1} stays within $1/7$ of 1 on this disk. Consequently

$$|t_1^q - t_2^q| \geq \frac{6q}{7}|t_1 - t_2| \quad (|t_i - 1| \leq s), \quad (3.6)$$

while (3.5) gives

$$|R(t_1) - R(t_2)| \leq (K_n/s)|t_1 - t_2|.$$

For large n , $K_n < qs/2$. If $G(t_1) = G(t_2) = 0$, comparison with (3.6) forces $t_1 = t_2$. Moreover

$$|qt^{q-1}| \geq 6q/7 > K_n/s,$$

so $G'(t) \neq 0$ at its zero. The zero is therefore unique and simple. This proves Proposition 1.3 by exactly the existence–uniqueness mechanism used in the formal development.

4. EXHAUSTION, DISJOINTNESS, AND COVERING

Fix r in a compact annulus and choose a summand of maximal modulus on $|z| = r$. The normalized block slopes satisfy

$$\frac{1}{q_j} \log \frac{|T_{j+1}(z)|}{|T_j(z)|} = \log r - \log \rho_j. \quad (4.1)$$

Because ρ_j is strictly increasing, these slopes strictly decrease with j . Hence the sparse terms increase to a single maximal region and then decrease.

Suppose r lies between two consecutive crossing radii. If r is outside fixed logarithmic neighborhoods of both crossings, the middle term dominates both neighbors; the tail estimate then makes it larger than the sum of all remaining terms. No zero can occur there. If r lies in one of the transition neighborhoods, the two-term estimate applies. At a zero, division by the dominant monomial and (3.4) force the phase to lie within $O(1/q_j)$ of an odd q_j -th root of -1 , hence the zero lies in a model disk. Since (2.9) keeps the relevant crossing radius in $[a/2, 2b]$, this proves Proposition 1.4.

Apply Proposition 1.3 on the enlarged annulus $A_{a/2, 2b}$ and choose its disk constant in Proposition 1.4. Every annular zero then lies in a model disk whose unique zero is simple. This proves Corollary 1.5.

For the disjointness assertion, model centers on one circle satisfy

$$|z_{j,\ell} - z_{j,\ell'}| \geq \frac{4\rho_j}{q_j} \quad (\ell \neq \ell'), \quad (4.2)$$

for all sufficiently large relevant j . With $c \leq 1$, disks on the same circle are therefore disjoint. Disks on different circles are separated by the radial gap (2.10), of order $n^{-1/6}$, whereas their radii are $O(q_j^{-1}) = O(n^{-1/3})$. Hence the whole admissible family is eventually pairwise disjoint, proving Proposition 1.6. Choosing the disk constant supplied by Proposition 1.3, disjointness

and uniqueness give one distinct simple zero per disk in any finite admissible family. This proves Proposition 1.7.

It remains to prove Theorem 1.2. Let $w \in A_{a,b}$ and put $r = |w|$. Choose j so that

$$\nu_j \leq n + r\sqrt{n} < \nu_{j+1},$$

with the first active index used in the harmless initial case. Equations (2.2) and (2.9) give

$$|\rho_j - r| \leq Cn^{-1/6}.$$

Choose the nearest model angle on the circle $|z| = \rho_j$. Its angular displacement from w is $O(q_j^{-1}) = O(n^{-1/3})$. Apply Proposition 1.3 on a fixed enlarged annulus containing all such crossing radii; it supplies a true zero within another $O(n^{-1/3})$. The radial error dominates, giving (1.6).

Finally let U be a nonempty open set. It contains a nonzero point w and a small closed disk about w contained in some annulus $A_{a,b}$. For all sufficiently large n , Theorem 1.2 places a zero of $F^{(n)}$ in that disk. This proves Theorem 1.1.

Remark 4.1 (The origin). The simplicity statement is deliberately annular. If $j_0 = \min\{j : \nu_j \geq n\}$, then the multiplicity of the zero at the origin is exactly $\nu_{j_0} - n$. For suitable derivative orders this multiplicity is large, so no global eventual-simplicity statement is true.

5. FORMAL VERIFICATION AND SCOPE

The Lean development is pinned to Lean/Mathlib 4.28.0. The main declarations corresponding to the statements above are:

Printed statement	Lean declaration
Entire function and derivative formula	<code>F_differentiable, iteratedDeriv_F</code>
Transcendence	<code>F_not_polynomial</code>
Upper growth bound (2.3)	<code>F_norm_le</code>
Theorem 1.2	<code>annular_covering_rate_p32</code>
Theorem 1.1	<code>erdos906_sparse_fock_p32</code>
Proposition 1.3	<code>model_disk_zero_count_one_p32</code>
Proposition 1.4	<code>annular_zero_exclusion_p32</code>
Corollary 1.5	<code>annular_zeros_simple_deriv_p32</code>
Proposition 1.6	<code>model_disks_pairwise_disjoint_p32</code>
Proposition 1.7	<code>zero_count_model_disk_union_p32</code>

The finite-union theorem uses ordinary cardinality of the set of distinct zeros; simplicity then makes this equal to total multiplicity on those disks. The published source contains no `sorry`, `admit`, custom `axiom`, `unsafe`, `native_decide`, or `@[implemented_by]` in the project modules. The main declarations are audited with `#print axioms`; only the standard dependencies `propext`, `Classical.choice`, and `Quot.sound` occur.

The present note is intentionally narrower than the accompanying research directory. Results there concerning general exponents, the endpoint $p = 4/3$, sectorial counting, limiting zero measures, and support-profile limits are not claims of this formalized submission unless and until they receive separate machine verification.

AI ASSISTANCE DISCLOSURE

Language-model tools were used substantially during derivation, checking, exposition, and formalization. The mathematical claims in this submission are separated from historical-priority claims, and the repository records the boundary of the machine-checked development.

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